

IMPERIAL

Backtest of ICES Advice

Historical, F_{target} , and Advice rule ---
one SRR branch

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Outline

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What was simulated

Six stocks, three trajectories, one operating-model branch

Stock	Area	Assessment	Advice basis
Irish Sea cod	7.a	Age-structured	ICES MSY
Irish Sea whiting	7.a	SAM	ICES MSY
Celtic Sea cod	7.e-k	SAM	ICES MSY
Celtic Sea whiting	7.b-c, 7.e-k	Age-based	ICES MSY
Celtic Sea haddock	7.b-k	SAM	ICES MSY
NE Atlantic mackerel	1-8, 14, 9.a	Integrated	Coastal States MSY

- **Historical** — latest WGCSE assessment.
- F_{target} — constant ICES F_{MSY} from 2005 (open-loop).
- **Advice rule** — closed-loop h_{crICES} .

Same OM biology and SRR residuals; only the harvest policy differs. **Historical and Future = bh3**; ICES HCR parameters from **bh1**. Not the full OM set.

Methods

Open loop = simple projection; closed loop = MSE with feedback

Open loop — F_{target}

Simple projection at constant F_{target} . No feedback.
fwd at 2005 SAG F_{MSY} from 2006 (lag 1).
Benchmark: fish at F_{MSY} every year, without the HCR
cutting F below $\text{MSYB}_{\text{trigger}}$.

Closed loop — Advice rule

MSE with feedback control: the ICES HCR revises F from SSB each year. hcrICES: $F = F_{\text{target}}$ when $\text{SSB} \geq \text{MSYB}_{\text{trigger}}$; reduced toward B_{lim} . True OM SSB ($\text{err} = \text{NULL}$); $\text{implErr} = 0$; lag 1.

They can disagree — Irish Sea cod historical $F \approx 0.07 \times F_{\text{target}}$.

Setting	This draft
Observation error	<code>err = NULL</code> (true OM SSB)
Implementation error	<code>implErr = 0</code>
Advice lag	1 year
Stochasticity	Deterministic (iteration 1)
HCR inputs	ICES SAG on bh1; held fixed
OM SRR	bh3 (R_0 prior; Historical and Future)

Why B_{MSY} comes from the OM

ICES does not report an MSY biomass target

- CFP target: restore stocks above B_{MSY} . ICES has no biomass target — F_{MSY} , B_{lim} , $MSYB_{trigger}$ only (Winker et al., Fish and Fisheries 2026).
- B_{MSY} is an implicit EqSim output (SSB at F_{MSY}) and is not published. $MSYB_{trigger}$ is usually B_{pa} ($\approx 1.4B_{lim}$), on average $\sim 60\%$ of B_{MSY} . The trigger is not a proxy for MSY biomass.
- Hence: ICES $MSYB_{trigger}$ parameterises the HCR; OM B_{MSY} (bh3 equilibrium) is the evaluation yardstick.
- EqSim as used in benchmarks is largely an **open-loop** projection at constant F . It is not MSE with feedback; EqSim F_{MSY} is not precautionary relative to a shortcut MSE on an OM (Winker et al., ICES JMS 2025, fsaf204).

Two Historical stories

bh3 trajectories; ICES $MSYB_{\text{trigger}} / F_{\text{target}}$ from bh1 --- not 24 ggplot panels

Irish Sea whiting — rebuild

Historical $F/F_{\text{target}} \approx 5.3$. Constant F_{target} (and AR once above trigger) cuts F and rebuilds SSB to $\sim 5 \times MSYB_{\text{trigger}}$.

The figure you would look at: Historical SSB stays below trigger; F_{target} and AR rise well above it.

Choke / recovery TAC versus an OM / F_{target} mismatch is not separated yet.

Irish Sea cod — counter-example

Historical $F/F_{\text{target}} \approx 0.07$. Fishing at F_{target} **lowers** terminal SSB versus history (AR $\Delta = -0.21$).

The figure you would look at: Historical F already far below the ICES line; putting the rule in force is a rise in F .

Current --- terminal year

SSB/MSYB_{trigger} and F/F_{target} --- Historical bh3, HCR inputs ICES/bh1

Stock	Hist SSB/MSYB _{trigger}	F_{target} SSB/MSYB _{trigger}	AR SSB/MSYB _{trigger}	Hist F/F_{target}	AR-Hist Δ SSB
Irish Sea cod	0.54	0.27	0.33	0.07	-0.21
Irish Sea whiting	0.51	4.94	4.95	5.25	+4.45
Celtic Sea cod	0.02	4.22	4.23	4.44	+4.21
Celtic Sea whiting	0.22	0.40	0.49	1.42	+0.27
Celtic Sea haddock	0.63	1.21	1.21	1.78	+0.58
NE Atlantic mackerel	0.77	0.77	0.77	1.39	0.00

All six below MSYB_{trigger} at the terminal year. Realised F exceeds F_{target} except Irish Sea cod. Under F_{target} , $F/F_{\text{target}} = 1$ by construction.

Three findings to take away

Historical and Future (bh3) --- this branch only

1. **Irish Sea cod.** Fishing at F_{target} lowers SSB versus history. Historical F is already far below the ICES target.
2. **Celtic Sea whiting.** ICES $MSYB_{\text{trigger}}$ sits above OM B_{MSY} . A 20-year HCR can reach MSY biomass and still be “below trigger”.
3. **Celtic Sea cod.** From history, $MSYB_{\text{trigger}}$ in ~ 9 years; OM B_{MSY} is not reached in 20 years from either start. The AR start is already above the trigger — different targets.

Future --- 20-year HCR

bh3, mean biology, mean recruitment; both starts then follow the ICES HCR

Stock	From Historical		From Advice rule	
	MSYB _{trigger}	B _{MSY}	MSYB _{trigger}	B _{MSY}
Irish Sea cod	8	6	8	7
Irish Sea whiting	2	7	0	0
Celtic Sea cod	9	—	0	—
Celtic Sea whiting	—	18	—	16
Celtic Sea haddock	4	—	0	—
NE Atlantic mackerel	3	0	3	0

Years to threshold; 0 = already above; — = not within 20 years. Reaching ICES MSYB_{trigger} does not imply OM B_{MSY}. Mackerel is already above B_{MSY} while still below the (SAG) trigger.

Caveats

Read the numbers as one branch, not a full MSE

- **bh3.** Historical and Future trajectories use bh3 (R_0 prior). HCR inputs (ICES F_{target} , $\text{MSYB}_{\text{trigger}}$, B_{lim}) stay at SAG values on bh1. OM shading (F_{MSY} , B_{MSY}) is bh3. bh1 is unconstrained Beverton-Holt — steepness is estimated, not 1.
- **Mackerel** $\text{MSYB}_{\text{trigger}}$. SAG field, equal to B_{pa} in the latest SAG row; not the LTMP / B_{lim} advice-sheet convention.
- **No estimation error.** $\text{err} = \text{NULL}$: the HCR sees true OM SSB. No implementation error; deterministic, not $P(\text{SSB} < B_{\text{lim}})$.

Next

What this draft does not yet do

1. **Other SRRs** — bh2 (FishLife steepness) and Ricker rk1–rk3, with HCR inputs still held at ICES values.
2. **Advised-catch series** — to separate a choke / recovery TAC from an OM / F_{target} mismatch (Irish Sea cod).
3. **Monte Carlo** $P(B_{\text{lim}})$ — stochastic replicates for Historical and Future, not a single deterministic trajectory.

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Questions / discussion

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